

Suggested FYP topics for 2026-27

by

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SCL1, SCL2, SCL3: Projected conducted in Prof. Liew's group in CUHK

SCL4, SCL5: Industry-sponsored projects. Main interactions with the company Zyetric.

Project code :	SCL1
Project title :	AI-Driven Medical Image Reconstruction for Reliable Remote Surgery Under Network Transmission Errors
No. of students :	1-2
Description of project :	Project Background: High-quality medical imaging is critical for the success of remote surgery, where doctors rely on real-time visual feedback to make precise decisions. However, during network transmission across different locations, issues such as packet loss and distortion can significantly degrade the quality of images and videos, impacting the surgeon's ability to operate effectively. Traditional error correction techniques and hardware redundancy methods can partially mitigate these issues but are insufficient for handling the unique distortions caused by medical image transmission in extreme network conditions. To address this, advanced AI-driven solutions offer a promising approach. Deep learning models, particularly those designed for image reconstruction, can leverage spatial and temporal features to restore distorted medical images, ensuring the accuracy and reliability of visual information during surgery. Detailed Problem Description: Despite advancements in AI and image reconstruction, remote surgery still faces challenges due to the unpredictable nature of network conditions. Packet loss during transmission often results in large-scale color blocks, asynchronous decoding, and other visual artifacts that are distinct from traditional image noise. These distortions make existing image processing techniques ineffective. Furthermore, the variability in network environments and image content requires models that are both robust and adaptable. The key challenge lies in designing a deep learning-based visual reconstruction framework that can handle these unique distortions while maintaining high computational efficiency and clinical reliability. This project aims to develop a novel AI-driven medical image enhancement system that leverages convolutional and Transformer-based architectures to recover high-quality surgical images, ensuring accurate and timely visual feedback for remote surgery applications.
Industrial collaboration :	No.

Project code :	SCL2
Project title :	Data-Driven Approaches for Integrated LiFi Sensing and Communication in Human-Computer Interaction
No. of students :	1-2
Description of project :	Project Background: Gesture recognition (GR) is an important research area in human–computer interaction (HCI), with applications ranging from healthcare assistance to smart homes and interactive services. Traditional GR methods based on cameras, acoustic sensing, or RF systems often suffer from privacy risks, environmental noise, and high computational requirements. Li-Fi technology, built on visible light communication, offers an attractive alternative due to its natural directionality, strong sensitivity to occlusion, and reduced susceptibility to RF interference. These physical properties make Li-Fi especially suitable for channel state information (CSI) sensing. By capturing and analyzing light signal variations caused by hand or finger movements, Li-Fi systems can achieve accurate and efficient gesture recognition. Beyond sensing, Li-Fi inherently serves as a high-speed wireless communication link, transmitting data through modulated light signals. This dual functionality naturally lends itself to the paradigm of integrated sensing and communication (ISAC), where the same Li-Fi infrastructure can simultaneously support data transmission and environmental perception without requiring additional hardware or dedicated spectrum resources. With low-cost LEDs and photodiodes serving as the hardware foundation, Li-Fi-based ISAC provides a scalable, non-intrusive, and energy-efficient platform for developing intelligent gesture recognition systems using modern data-driven approaches, while maintaining seamless communication capabilities. Detailed Problem Description: Despite rapid progress in gesture recognition technologies, real-world deployment still faces significant challenges. Camera-based methods raise privacy concerns and require heavy computation; acoustic and RF-based systems are vulnerable to multipath fading, background noise, or limited resolution; while radar and depth sensors demand costly, specialized hardware. These limitations highlight the need for a robust and low-cost alternative. Li-Fi-based gesture recognition addresses this need by leveraging the sensitivity of visible light signals to interruptions caused by hand or finger movements. The resulting CSI variations form rich, multi-dimensional data that can be processed and classified using a range of machine learning and deep learning techniques. Depending on how the signal is represented, suitable approaches may include, but are not limited to, recurrent architectures (e.g., LSTM) for sequential

	modelling, convolutional neural networks (CNNs) for spectrogram or image-based feature extraction, and fully connected deep neural networks (DNNs) for learning compact discriminative representations. Hybrid or attention-based architectures may also be explored to capture both temporal dynamics and cross-channel correlations. The key challenge lies in designing effective signal representations, model architectures, and training strategies that can adapt to diverse users, gesture speeds, and illumination conditions while maintaining reliable recognition accuracy.
Industrial collaboration :	No.

Project code :	SCL3
Project title :	Construction of a Semantics-Annotated Vehicular Image Dataset for Efficient Semantic Communication
No. of students :	1-2
Description of project :	Project Background: Semantic communication is an emerging paradigm that transmits only the most meaningful information from a source, offering significant improvements in bandwidth efficiency and communication latency over conventional methods. In vehicular networks, where real-time and reliable data exchange is essential for safety-critical applications such as autonomous driving and traffic monitoring, semantic communication holds particular promise. A key enabler of this paradigm is Joint Source-Channel Coding (JSCC), which jointly optimizes source compression and channel protection to maximize transmission efficiency. Recent research has shown that integrating Multimodal Large Language Models (MLLMs) into JSCC encoders can further enhance performance by enabling the system to assess the semantic significance of different regions within a transmitted image — for example, distinguishing safety-critical objects like nearby vehicles and pedestrians from less important background elements such as the sky or distant buildings. By assigning variable coding lengths based on semantic importance, MLLM-assisted JSCC encoders can substantially reduce bandwidth usage without sacrificing critical traffic-related information. However, the development and evaluation of such semantics-aware encoders is currently bottlenecked by the lack of publicly available vehicular image datasets annotated with object-level semantic importance labels tailored to transportation contexts. This project aims to fill this gap by constructing a large-scale, well-annotated vehicular image dataset that can serve as a foundational resource for research in semantics-aware vehicular communication. Detailed Problem Description: Existing large-scale driving datasets, such as Cityscapes and BDD100K, provide rich collections of urban street-scene images with annotations for tasks like object detection and semantic segmentation. However, these annotations do not capture the contextual importance of objects

	from a vehicular communication perspective — for instance, a nearby pedestrian crossing the road is far more critical to driving safety than a distant parked car, yet both may share the same segmentation label in current datasets. Bridging this gap requires a systematic re-annotation effort that assigns multi-level importance labels (e.g., high, medium, low) to objects and image regions based on their relevance to driving safety and traffic awareness. This project aims to develop such an annotated dataset by leveraging LLM-assisted data labeling pipelines applied to existing driving datasets. Students will design and iteratively refine prompting strategies for LLMs (e.g., GPT-4o or comparable models) to automatically generate semantic importance labels for objects and image patches, followed by human review and quality control to ensure annotation consistency and reliability. The key challenges include designing effective prompting templates that capture the nuanced priorities of diverse vehicular scenarios (e.g., highway vs. urban intersection), establishing robust and reproducible annotation criteria, handling ambiguous cases where object importance is context-dependent, and validating the quality and utility of the resulting dataset through statistical analysis and preliminary downstream evaluations. The final deliverable will be a publicly releasable, well-documented dataset that supports future research in semantics-aware image communication for intelligent transportation systems.
Industrial collaboration :	No.

Project code :	SCL4
Project title :	Cost reduction for AI training and inferencing
No. of students :	2
Description of project :	Introduction: Currently most AI related businesses focus on “being the first to launch some AI models”, therefore most businesses are fighting for the latest and fastest GPU resources on their R&D projects. This trend is causing the GPU costs, especially the cloud based GPU rentals, to skyrocket, and impacting significantly to SMEs like our company. To reduce the overall GPU costs for Zyetric Technologies, we need to research on the following topics: 1. How to make the the AI models more cost efficient in both training and inferencing in the field of facial translation (e.g., popular AI models such as LivePortrait)? 2. How to optimize the such AI models to run on older and cheaper GPUs but maintain the similar performance? Technical Challenge: 1. Need to understand the fundamentals of AI model training and inferencing in the field of facial translation very well;

	2. Need to understand the source code of popular AI related frameworks, such as Pytorch and Tenserflow, very well. Benefits and Technical Know-how you will acquire: You will get a deep understanding of the fundamentals of AI model training and inferencing; You will be well prepared for AI related R&D jobs after graduation.
Industrial collaboration :	Yes, by Zyetric Technologies Limited. You will be supervised and guided by senior scientists at Zyetric. You will need to be interviewed by Zyetric before being accepted into this project. If you are interested in this project, please write an email to to Prof. Soung Liew (soung@ie.cuhk.edu.hk). Prof. Liew will then link you up with Zyetric for an interview. You will be accepted into this project if there is a match: i.e., you are comfortable with them and they are comfortable with you. The interview process is a prerequisite.

Project code :	SCL5
Project title :	Reducing or eliminating the hallucination problem in AI inferencing
No. of students :	2
Description of project :	Introduction: Hallucination is one of the biggest problems in AI inferencing, which significantly limits the usability of many LLMs in many areas such as AI translation. In this project we will focus on reducing or eliminating hallucination in AI translation, we need to research on the following topics: 1. What is the main source of hallucination in LLMs? 2. What are the current solutions for hallucination? 3. How can we apply those solutions onto AI translation? Technical Challenge: 1. Need to test the current solutions for hallucination in some translation focusing LLMs such as Alicloud's Bailian model and see which one makes more sense; 2. Might need to think outside the box and even invent some solutions for the problem. Benefits and Technical Know-how you will acquire: You will get a deep understanding of the fundamentals of the hallucination problem in AI model inferencing; You will be well prepared for AI related R&D jobs after graduation.
Industrial collaboration :	Yes, by Zyetric Technologies Limited.

	You will be supervised and guided by senior scientists at Zyetric. You will need to be interviewed by Zyetric before being accepted into this project. If you are interested in this project, please write an email to Prof. Soung Liew (soung@ie.cuhk.edu.hk). Prof. Liew will then link you up with Zyetric for an interview. You will be accepted into this project if there is a match: i.e., you are comfortable with them and they are comfortable with you. The interview process is a prerequisite.
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